bitswarm

THE CONTROL PLANE FOR PHYSICAL AI

PRE-SEED

BitSwarm
Kubernetes for physical AI.

Across robots and on site devices.

K8S NODE	WORKLOAD	SCHEDULER	CLUSTER STATE
robot	**mission**	**BitSwarm**	**world model**

THE PROBLEM

Robot ready sites hardly scale up

Robots can do useful work	But still operate alone
No single robot does every job	Sites must combine vendors
Each robot needs an operator	Costs grow with headcount
Each robot keeps its own data	No shared site memory

TRADITIONAL

Cloud base

One pilot per robot

Robots (UGV / UAV)

Single link, no relay, no shared memory.

FIELD RESEARCH

54 questionnaires · **15** potential customers · **4** deep-dive interviews · **2** US design partner

32.1%

cite signal loss and unstable connectivity as their top field challenge

57.2%

struggle with siloed data, repeated setup, and no unified dashboard

WHY NOW

The hardware has arrived. The intelligence hasn't.

	INDIVIDUAL INTELLIGENCE one smart robot	SWARM INTELLIGENCE one smart fleet
Where the intelligence lives	Inside each machine	In the coordination layer
When one robot fails	The mission stops	The fleet reroutes and continues
To scale a deployment	Wait for smarter robots	Add another robot
Ready	**Years out**	**Buildable today**

THE SOLUTION

One execution layer across robots and vendors

01 Connect every robot

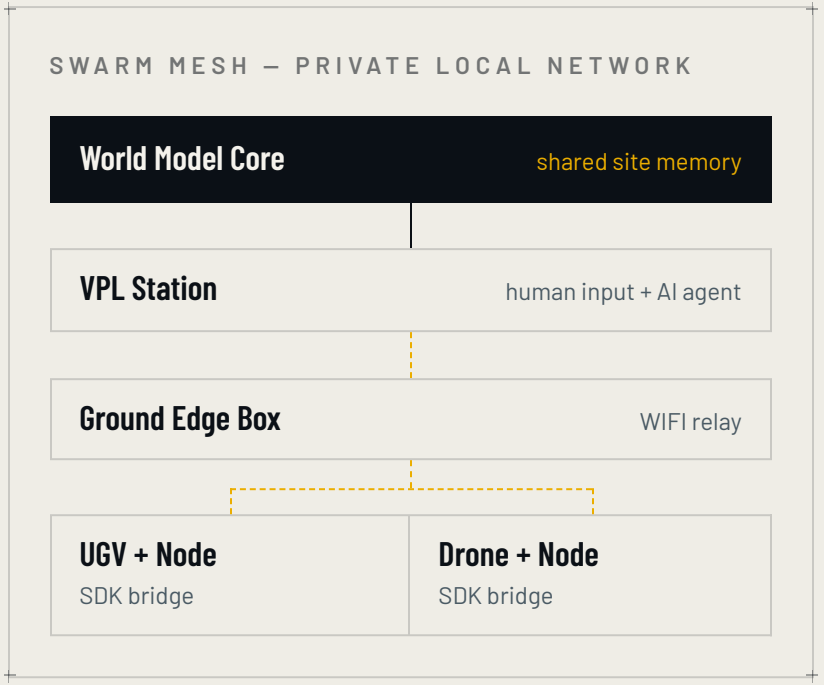
Bridge different robot links and SDKs through one edge layer.

02 Coordinate the fleet

One control plane assigns and supervises missions across robots.

03 Build shared memory

Every mission updates the same world model for the site.



THE WEDGE

Two design partners define the wedge

DESIGN PARTNER 01

Construction — VDC team

~~10h+~~ **2h**

Walking the site becomes desk review — BitSwarm runs the field collection.

Spend vs. classical solution	**Nearly halved**
Field efficiency	**Super-linear scalable**

DESIGN PARTNER 02

Drone services — cleaning · inspection · mapping

Their own task memory

Every job becomes a simple, repeatable workflow — with a human in the loop monitoring.

Customization per job type	**Once**
Following task execution	**Automate from memory**

BUSINESS MODEL

Robot as a Service

01 SOFTWARE — CONTROL PLANE

PER SITE, PER YEAR · TIERED BY COORDINATED HARDWARE (ROBOTS + FIXED CAMS)

UP TO 5 DEVICES	UP TO 10	UP TO 15	15+
$10K /site/yr	$16K /site/yr	$25K /site/yr	Custom

02 ROBOT RENTAL (INCLUDE SIGNAL RELY NODE)

$700–1,500 /month

Depending on robot type.

03 CLOUD SERVICE

Usage-based

Price follows consumption.

DEVELOPMENT PLAN

Delivered in this round

NOW

Live with design partners

The Node bridges, quadrupeds, drones and fixed cams; supervised missions run on real sites.

2 DESIGN PARTNERS

+4-6 MONTHS

Control plane v1

Multi-robot scheduling with a human in the loop; design partners convert to paying sites.

FIRST PAYING SITES

+8–12 MONTHS

Task memory across sites

The shared world model ships site-wide; the rental fleet runs day to day.

3+ PAID SITES

SEED GATE

Ready to scale

Deeply bind with customer habbits, proving the model compounds with every mission.

5+ SITES · $200K+ ARR

MARKET SIZE

Every site that runs without people on it

The system runs on any site where mixed robots do repeatable field work — **only the mission library changes.**

CONSTRUCTION — CURRENT WEDGE **A site that keeps building 24/7 — no human on it overnight**	HARBOR & PORTS **A terminal that inspects itself between ships**	MINING FIELDS **Nobody walks into the hazardous zone again**
ENERGY **Assets that never wait for an inspection window**	FIRST RESPONDERS **The swarm is airborne before the crew arrives**	SCHOOLS & CAMPUSES **Grounds that patrol themselves, night and day**

COMPETITION LANDSCAPE

DIMENSION	DroneDeploy capture & analytics	Field AI per-robot autonomy	BitSwarm control plane
One operator with multiple robots	—	■■	■■■
Offline / harsh-site connectivity	—	■■	■■■
Live shared site state	■	■■	■■■
Mission orchestration across a fleet	■	■■	■■■
No OEM lock-in	■	■	■■■
Scalability & Cost	■■	■	■■■
Individual intelligence	■	■■■	■■

■ capability depth — — none

TEAM



FOUNDER

Zirui (Bill) Wang

Founder & CEO

CALTECH

PhD program, Computational Mathematics — mastered out to build BitSwarm

NCSA

Research assistant, National Center for Supercomputing Applications two years experiences